

Direct Robust Function-Private Identity-Based Encryption from Lattices

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Abstract. Function privacy for identity-based encryption (IBE) requires that a secret key reveal no unnecessary information about the identity for which it was generated. Boneh, Raghunathan, and Segev introduced this notion and constructed function-private anonymous IBE schemes, while observing that robustness can be obtained by applying the generic transformations of Abdalla, Bellare, and Neven. However, directly constructing robust function-private anonymous IBE, especially from lattices, remained a gap. In this work, we construct a standard-model LWE-based function-private anonymous IBE scheme satisfying strong robustness without applying the ABN generic transformations. Our construction follows the ABB lattice IBE framework, but incorporates an internal identity-binding check based on full-rank-difference (FRD) encoding, gadget-opening verification, and universal hashing over $\mathcal{R}_q$. The scheme achieves statistical function privacy for high-min-entropy identity sources, anonymity under the LWE assumption, and strong robustness from the uniqueness of LWE and gadget openings together with a q^{-d} hash-collision bound. Compared with the BRS13 standard-model LWE construction, our scheme removes the $\mu = \omega(\log \lambda)$ repetition layer and gives asymptotically smaller identity-dependent public parameters, ciphertext components, and secret-key seed components.

Keywords: Identity-based encryption · Anonymity · Function privacy · Robust encryption · Lattice-based cryptography · Learning with errors

1 Introduction

Identity-based encryption (IBE) is a public-key encryption primitive in which the decryption key sk_{id} is generated from a receiver identity id [5,1,6]. In a standard IBE scheme, anyone can encrypt a message associated with an identity id , while only the holder of the corresponding secret key sk_{id} can decrypt.

In many IBE applications, however, the identity used for encryption may itself be sensitive. For example, in public-key searchable encryption, an identity

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may encode a receiver, a keyword, or some other search-related attribute. Revealing this value from the ciphertext may already leak sensitive information. This motivates anonymous IBE, also called recipient-anonymous IBE, where ciphertexts hide not only the encrypted message but also the identity under which the ciphertext was generated [9,1,6]. In the standard anonymity experiment, an adversary should not be able to distinguish whether a challenge ciphertext was generated under one identity or another.

Ciphertext anonymity alone does not protect all identity information in an IBE system. If a secret key sk_{id} reveals its associated identity id , then a key holder can use this information to test ciphertexts associated with the same identity. A successful decryption indicates that the ciphertext was intended for that identity. Thus, to hide identities from both ciphertexts and secret keys, one needs a key-side privacy notion. Boneh et al. introduced this notion as function privacy for IBE and constructed the first function-private anonymous IBE schemes [6]. Function privacy requires that a secret key reveal no unnecessary information about its corresponding identity. Since a secret key can always be tested against ciphertexts encrypted under guessed identities, this notion is meaningful only when identities have sufficient entropy. BRS13 formalised this requirement using unpredictable identity sources and showed that statistical function privacy can be achieved together with anonymity.

Robustness for Function-Private Anonymous IBE. Robustness, formalised by Abdalla et al. [1], requires that it should be infeasible to produce a ciphertext that is accepted under two different identities. This requirement is closely related to anonymous encryption. In an anonymous encryption scheme, the ciphertext does not reveal the identity under which it was generated. Hence, a receiver cannot decide from the ciphertext alone whether it is the intended recipient. This decision must be made by decryption. If the same ciphertext can be accepted under two different identities, then the intended recipient of the ciphertext is not well defined.

This issue is particularly relevant to function-private anonymous IBE. In this setting, ciphertexts should hide the encryption identity and secret keys should hide the key identity. Therefore, identity consistency cannot be enforced by explicitly combining the identity in the ciphertext or in the secret key. It has to be guaranteed by the cryptographic structure of the scheme. Abdalla et al. [1] introduced weak and strong robustness notions and gave generic transformations that convert suitable encryption schemes into robust ones. Since strong robustness implies weak robustness [1], in this work we focus on strong robustness. Boneh et al. [6] also mentioned that the ABN10 generic transformations can be applied to their function-private IBE schemes because they do not change the decryption keys.

However, this only gives robustness through an external generic transformation. Boneh et al. give a standard-model LWE construction following an extract–augment–combine approach. To make wrong-identity acceptance negligible when q is polynomial, their construction uses $\mu = \omega(\log \lambda)$ parallel copies together with bit decomposition. As a result, the identity-dependent public parameters

Table 1. Compact comparison with the BRS13 standard-model LWE construction [6]. We count explicit $\mathbb{Z}_q$ entries in the public parameters, ciphertexts, and the seed/hash part of secret keys, excluding fixed structured matrices such as $\mathbf{G}$ and lattice basis objects.

Scheme	PP	CT	SK seed/hash	Robust
BRS13 LWE2 [6]	$\Theta((\ell\mu d + 2)nm)$	$\Theta(m + \ell\mu d m + 1)$	$\Theta(\ell\mu d)$	no
Ours	$\Theta(nm + \ell n w)$	$\Theta(m + \ell w + 1)$	$\Theta((\ell + 1)d)$	yes

and ciphertext blocks are indexed by $(i, j, k) \in [\ell] \times [\mu] \times [d]$. Since this lattice construction is not directly robust, Boneh et al. [6] explicitly leave directly constructing robust function-private IBE schemes without relying on the ABN10 generic transformations as an open problem.

1.1 Our Contributions

We construct a compact standard-model LWE-based function-private anonymous IBE scheme satisfying strong robustness. Our construction addresses the direct-robustness problem left open by BRS13 in the lattice setting. It does not apply the ABN10 generic transformations. Instead, the identity-binding mechanism is built into the LWE and gadget-opening structure of the scheme.

The construction achieves statistical function privacy using the same leftover-hash principle as BRS13, anonymity under LWE, and strong robustness from the uniqueness of the relevant LWE and gadget openings. The main technical difference from the BRS13 standard-model LWE construction is that we remove the μ -fold repetition layer. Boneh et al. use $\mu = \omega(\log \lambda)$ repetitions to make wrong-identity acceptance negligible when q is polynomial. Our construction instead uses extended full-rank-difference encoding (FRD) and gadget-opening verification to obtain a direct identity-binding check.

Table 1 shows that our construction removes the $\mu = \omega(\log \lambda)$ repetition layer used in BRS13. In particular, the number of identity-dependent public matrices drops from $\ell\mu d$ to ℓ , and the number of identity-dependent ciphertext blocks drops from $\ell\mu d$ to ℓ . In the secret key, although $\mathbf{Y}$ is written as a matrix for notation, it is stored by its ring representation $y \in \mathbb{Z}_q^d$. Hence the seed/hash part $(\mathbf{s}, \mathbf{Y})$ has size $\Theta((\ell + 1)d)$. The full decryption key additionally contains the lattice basis $\mathbf{T}_{\mathbf{s}, \mathbf{Y}}$ for $\Lambda_q^\perp(\mathbf{F}_{\mathbf{s}, \mathbf{Y}})$, which is not included in the seed/hash column. Our construction also provides strong robustness internally, whereas BRS13 LWE2 is not robust.

1.2 Technical overview.

The main role of the extended FRD encoding is to replace the repetition-based identity compression of BRS13 by a compact compression with collision probability q^{-d} . A direct compression over $\mathbb{Z}_q$ would only give a collision probability

about $1/q$, which is not negligible when q is polynomial. Boneh et al., therefore use $\mu = \omega(\log \lambda)$ repetitions to amplify this probability. We instead work over the degree- d ring $\mathcal{R}_q$, whose size is q^d , and use the extended FRD map to embed ring elements into $n \times n$ matrices. For an identity $\text{id} = (\text{id}_1, \dots, \text{id}_\ell)$, key generation samples a fresh seed $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_\ell) \leftarrow U(\mathcal{R}_q^\ell)$ and computes $\mathbf{Y} = \sum_{i=1}^\ell H(\text{id}_i)H(\mathbf{s}_i) \in \mathbb{Z}_q^{n \times n}$. Although $\mathbf{Y}$ is matrix-valued, it is the FRD image of the ring compression $\sum_i \text{id}_i \mathbf{s}_i$, since the extended FRD map H is a ring homomorphism. Thus its collision probability is still bounded by q^{-d} . The matrix $\mathbf{Y}$ is then embedded directly into the lattice block as $\mathbf{Y}\mathbf{G}$.

This compression serves two purposes. First, it gives function privacy. The family $f_{\mathbf{s}}(\text{id}) = \sum_{i=1}^\ell H(\text{id}_i)H(\mathbf{s}_i)$ is universal with collision probability at most q^{-d} . Hence, by the leftover-hash lemma, the pair $(\mathbf{s}, \mathbf{Y})$ statistically hides high-min-entropy identities. Second, it makes wrong-identity acceptance negligible. If a key for id is tested against a ciphertext structure associated with another identity id' , the relevant difference is $\sum_{i=1}^\ell H(\text{id}_i - \text{id}'_i)H(\mathbf{s}_i) = H\left(\sum_{i=1}^\ell (\text{id}_i - \text{id}'_i)\mathbf{s}_i\right)$. For distinct identities, this value is zero with probability at most q^{-d} . Hence, a randomly generated key can decrypt a ciphertext for a different identity only with negligible probability.

The same encoding also enables identity recovery during decryption. After the LWE randomness is recovered, each identity block has the form $\mathbf{G}^\top H(\text{id}_i)^\top \mathbf{r}$ plus a small error. The gadget inversion step removes the error and recovers $H(\text{id}_i)^\top \mathbf{r}$. Since $\mathbf{r} \neq \mathbf{0}$ and the map $\text{id}_i \mapsto H(\text{id}_i)^\top \mathbf{r}$ is injective, recovering id_i is equivalent to solving the corresponding determined linear system. The decryption algorithm then checks whether the recovered identity is consistent with the matrix value $\mathbf{Y}$ stored in the secret key, namely, whether $\mathbf{Y} = \sum_{i=1}^\ell H(\widehat{\text{id}}_i)H(\mathbf{s}_i)$.

Finally, strong robustness follows from unique opening. The LWE part of the ciphertext can be viewed as a commitment to the public-key randomness $\mathbf{r}$. Under the chosen parameters, an accepting ciphertext has a unique LWE opening, and the gadget blocks have unique identity-dependent openings. Hence, a ciphertext that decrypts under two distinct identities must recover the same claimed identity in both decryptions. Acceptance under a key for $\text{id}^{(b)}$ then forces $\sum_{i=1}^\ell H(\text{id}_i^{(b)})H(\mathbf{s}_{b,i}) = \sum_{i=1}^\ell H(\widehat{\text{id}}_i)H(\mathbf{s}_{b,i})$. For at least one of the two distinct identities, this is a non-trivial collision in the universal family above, which occurs with probability at most q^{-d} . This gives an internal SROB without the ABN generic transform.

1.3 Related Work

Function privacy has been studied mainly in identity-based encryption, predicate encryption, attribute-based encryption, and inner-product functional encryption. Boneh, Raghunathan, and Segev introduced function privacy for IBE and constructed the first function-private anonymous IBE schemes [6]. They later extended the framework to subspace-membership encryption and showed how to obtain function-private encryption schemes from inner-product encryption [7]. Han and Qin considered function privacy in attribute-based encryption

and applied it to searchable encryption [11], while Tang studied forward-security properties for PEKS and IBE, including forward-secure function privacy [18]. In the private-key functional-encryption setting, Bishop, Jain, and Kowalczyk introduced function-hiding inner-product encryption [4], and subsequent works improved its practicality and efficiency [12,13]. Patranabis, Mukhopadhyay, and Ramanna studied public-key function-private predicate encryption for low-min-entropy predicates, including zero and non-zero inner-product predicates [17]. Bartusek et al. constructed public-key function-private hidden vector encryption and more general function-private predicate encryption for small-superset predicates [3]. More recent work further considers function privacy in attribute-based encryption and efficient function-hiding inner-product functional encryption [15,19]. These works address function privacy for broader predicate or functional-encryption classes, whereas our work focuses on directly constructing a robust function-private anonymous IBE scheme from LWE.

2 Preliminaries

We use $\lambda \in \mathbb{N}$ for the security parameter. A function $\text{negl}(\lambda)$ is negligible if it is smaller than any inverse polynomial of λ . For a finite set S , we write $U(S)$ for the uniform distribution over S and $x \leftarrow U(S)$ to denote that x is sampled uniformly at random from S . For a positive integer ℓ , let $[\ell] = \{1, \dots, \ell\}$.

Vectors are column vectors by default. We use lowercase bold letters for vectors, e.g., $\mathbf{x}$, and uppercase bold letters for matrices, e.g., $\mathbf{A}$. “ $\mathbf{T}$ ” denotes vector transpose.

For $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$, we use the Euclidean norm $\|\mathbf{x}\|_2 = (\sum_i x_i^2)^{1/2}$ and infinity norm $\|\mathbf{x}\|_\infty = \max_i |x_i|$. For a matrix $\mathbf{X}$, $\|\mathbf{X}\|$ denotes the maximum ℓ_2 norm of its columns. For vectors over $\mathbb{Z}_q$, norms are taken on centred representatives in $(-q/2, q/2]$.

$\Delta(D_0, D_1)$ denotes the statistical distance between distributions D_0, D_1 . We say D_0 and D_1 are statistically close if $\Delta(D_0, D_1)$ is negligible in the security parameter.

2.1 Min-Entropy and Universal Hashing

For a random variable X over a finite set, its min-entropy is

$$H_\infty(X) = -\log \left(\max_x \Pr[X = x] \right).$$

A random variable X is a k -source if $H_\infty(X) \geq k$.

A family of functions $\mathcal{H} = \{h : U \rightarrow V\}$ is universal if, for all distinct $x, x' \in U$,

$$\Pr_{h \leftarrow U(\mathcal{H})} [h(x) = h(x')] \leq \frac{1}{|V|}.$$

A joint distribution $X = (X_1, \dots, X_T)$ over U^T is a (T, k) -block-source if, for every $i \in [T]$ and every fixing $x_1, \dots, x_{i-1}$ of the previous blocks, the conditional distribution $X_i \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}$ is a k -source.

Lemma 1 (Block-source leftover hash lemma [6, Lemma 2.3]). *Let $\mathcal{H} = \{h : U \rightarrow V\}$ be a universal hash family, and let $(X_1, \dots, X_t)$ be a (t, k) -block source over U^t . If $k \geq \log |V| + 2\log(1/\varepsilon) + \Theta(1)$, then the distribution $(h_1, h_1(X_1), \dots, h_t, h_t(X_t))$, $(h_1, \dots, h_t) \leftarrow U(\mathcal{H}^t)$, is εt -close to the uniform distribution over $(\mathcal{H} \times V)^t$.*

In our construction, the source domain is $U = \mathcal{R}_q^\ell$. The extracted value is represented as an $n \times n$ matrix, but since H is a ring homomorphism, it has the form $H(y)$ for some $y \in \mathcal{R}_q$. Hence the effective range has size $|\mathcal{R}_q| = q^d$. Thus, the entropy threshold in the block-source leftover hash lemma becomes $\log |\mathcal{R}_q| + 2\log(1/\varepsilon) + \Theta(1) = d\log q + 2\log(1/\varepsilon) + \Theta(1)$.

2.2 Lattices, LWE and Trapdoors

We use standard q -ary lattice notation and discrete Gaussian distributions. For a prime modulus q , a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, and a vector $\mathbf{u} \in \mathbb{Z}_q^n$, define the q -ary random integer lattice $\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{e} \in \mathbb{Z}^m : \mathbf{A}\mathbf{e} = \mathbf{0} \bmod q\}$, and its shift $\Lambda_q^\mathbf{u}(\mathbf{A}) = \{\mathbf{e} \in \mathbb{Z}^m : \mathbf{A}\mathbf{e} = \mathbf{u} \bmod q\}$. For a lattice or coset $L \subseteq \mathbb{Z}^m$ and $\sigma > 0$, define the density function $\rho_{\sigma, \mathbf{c}}(\mathbf{x}) = \exp\left(-\pi \frac{\|\mathbf{x} - \mathbf{c}\|^2}{\sigma^2}\right)$. The discrete Gaussian distribution over L with centre $\mathbf{0}$ and parameter σ is $D_{L, \sigma}(\mathbf{x}) = \frac{\rho_{\sigma, \mathbf{c}}(\mathbf{x})}{\sum_{\mathbf{y} \in L} \rho_{\sigma, \mathbf{c}}(\mathbf{y})}$ for $\mathbf{x} \in L$.

The security of our construction is based on the following decisional LWE assumption.

Definition 1. *Let q be prime, n, m be positive integers, χ be an error distribution over $\mathbb{Z}_q$. The advantage of an adversary $\mathcal{A}$ for the $\text{LWE}_{n, m, q, \chi}$ problem,*

$$\text{Adv}_{\mathcal{A}}^{\text{LWE}_{n, m, q, \chi}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{A}^\top \mathbf{r} + \mathbf{e}) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1]|$$

is negligible in λ where $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, $\mathbf{r} \leftarrow U(\mathbb{Z}_q^n)$, $\mathbf{e} \leftarrow \chi^m$, $\mathbf{b} \leftarrow U(\mathbb{Z}_q^m)$.

We use the gadget trapdoors of [16]. Let $q > 2$ be prime, $k = \lceil \log_2 q \rceil$, and $\mathbf{g}^\top = (1, 2, \dots, 2^{k-1}) \in \mathbb{Z}_q^{1 \times k}$. Set $w = nk$ and define the standard gadget matrix $\mathbf{G} := \mathbf{I}_n \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{n \times w}$. Let $m \geq 2n \log q + \omega(\sqrt{\log n})$ and $r = \omega(\sqrt{\log n})$. Define its minimum gadget distance as $\delta_{\mathbf{G}} := \min_{\Delta \in \mathbb{Z}_q^n \setminus \{\mathbf{0}\}} \|\langle \Delta^\top \mathbf{G} \rangle_q\|_\infty$, where $\langle \cdot \rangle_q$ denotes the centred representative in $(-q/2, q/2]$. For the standard gadget above, $q/4 \leq \delta_{\mathbf{G}} \leq q/2$: indeed, for any nonzero centred coordinate a , repeated doubling reaches magnitude at least $q/4$ before its magnitude reaches $q/2$, and the corresponding block $(a, 2a, \dots, 2^{k-1}a)$ occurs in $\Delta^\top \mathbf{G}$.

We recall the following lattice trapdoor generation, delegation algorithms and Gaussian sampling algorithms developed in [16].

Lemma 2 ([16]). *Let $r \geq \omega(\sqrt{\log n})$, $\bar{\mathbf{A}} \leftarrow U(\mathbb{Z}_q^{n \times (m-w)})$, and let $\mathbf{H} \in \mathbb{Z}_q^{n \times n}$ be invertible. There is a p.p.t algorithm $\text{TrapGen}(\bar{\mathbf{A}}, \mathbf{H}, r)$ that returns a matrix $\mathbf{A} = [\bar{\mathbf{A}} | \bar{\mathbf{A}}\mathbf{R} + \mathbf{H}\mathbf{G}] \in \mathbb{Z}_q^{n \times m}$ and a trapdoor $\mathbf{R} \in \mathbb{Z}^{(m-w) \times w}$ such that $\mathbf{A}$ is statistically close to $U(\mathbb{Z}_q^{n \times m})$ and $\mathbf{R}$ is statistically close to $D_{\mathbb{Z}, r}^{(m-w) \times w}$.*

Lemma 3 ([10]). *Let q be prime and $m \geq 2n \log q$ (in particular, $m \geq n$). For $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$ and $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, s}$ with $s \geq \omega(\sqrt{\log m})$, the joint distribution $(\mathbf{A}, \mathbf{A}\mathbf{e} \bmod q)$ is statistically close to $(\mathbf{A}, \mathbf{u})$, where $\mathbf{u} \leftarrow U(\mathbb{Z}_q^n)$. Furthermore, for every fixed full-row-rank $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{u} \in \mathbb{Z}_q^n$, the distribution of $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, s}$ conditioned on $\mathbf{A}\mathbf{e} = \mathbf{u} \bmod q$ is $D_{A_q^\mathbf{u}(\mathbf{A}), s}$.*

We also use the following extended leftover-hash lemma to analyse the matrix distributions arising in the anonymity proof.

Lemma 4 (Extended Leftover Hash Lemma, [2]). *Let $q > 2$ be prime, let $k = \text{poly}(n)$, and suppose $m > (n+1) \log q + \omega(\log n)$. For every fixed $\mathbf{v} \in \mathbb{Z}_q^m$, the distributions*

$$(\mathbf{A}, \mathbf{A}\mathbf{R}, \mathbf{R}^\top \mathbf{v}) \quad \text{and} \quad (\mathbf{A}, \mathbf{B}, \mathbf{R}^\top \mathbf{v})$$

are statistically close, where $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, $\mathbf{B} \leftarrow U(\mathbb{Z}_q^{n \times k})$, and $\mathbf{R} \leftarrow U(\{-1, 1\}^{m \times k})$ are sampled independently.

The following lemma derives the joint form required in our anonymity proof, where the matrix product and the correlated vector product must be statistically hidden simultaneously.

Lemma 5 (Correlated matrix leftover hashing). *Let $q > 2$ be prime, let $K = \text{poly}(n)$, and suppose $m > (n+2) \log q + \omega(\log n)$. Sample independently $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, $\mathbf{v} \leftarrow U(\mathbb{Z}_q^m)$, and $\mathbf{R} \leftarrow U(\{-1, 1\}^{m \times K})$. Then*

$$(\mathbf{A}, \mathbf{v}, \mathbf{A}\mathbf{R}, \mathbf{R}^\top \mathbf{v}) \approx_s (\mathbf{A}, \mathbf{v}, \mathbf{B}, \mathbf{z}),$$

where $\mathbf{B} \leftarrow U(\mathbb{Z}_q^{n \times K})$ and $\mathbf{z} \leftarrow U(\mathbb{Z}_q^K)$ are mutually independent and independent of $(\mathbf{A}, \mathbf{v})$.

Proof. Define the stacked matrix $\tilde{\mathbf{A}} = [\mathbf{A}^\top \mid \mathbf{v}]^\top \in \mathbb{Z}_q^{(n+1) \times m}$. Since $\mathbf{A}$ and $\mathbf{v}$ are independent and uniform, $\tilde{\mathbf{A}}$ is uniform, and

$$\tilde{\mathbf{A}}\mathbf{R} = \begin{bmatrix} \mathbf{A}\mathbf{R} \\ \mathbf{v}^\top \mathbf{R} \end{bmatrix}.$$

Apply Lemma 4 with row dimension $n+1$ and zero auxiliary leakage. Its uniform output splits into the independent pair $(\mathbf{B}, \mathbf{z}^\top)$, proving the claimed joint statistical closeness.

Lattice trapdoors allow solving the LWE problem efficiently. The uniqueness of the LWE problem solutions is crucial for proving the robustness of our IBE scheme. The following lemmas demonstrate this.

Lemma 6 (Uniqueness of gadget-LWE openings). *Let $q > 2$ be prime and let $\mathbf{G} \in \mathbb{Z}_q^{n \times w}$ be the standard gadget matrix defined above. Suppose $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{G} + \mathbf{e}^\top \pmod{q}$ with $\|\mathbf{e}\|_\infty < \delta_{\mathbf{G}}/2$. Then this opening is unique. Namely, if also $\mathbf{b}^\top = \mathbf{s}'^\top \mathbf{G} + \mathbf{e}'^\top \pmod{q}$ with $\|\mathbf{e}'\|_\infty < \delta_{\mathbf{G}}/2$, then $\mathbf{s} = \mathbf{s}'$ and $\mathbf{e} = \mathbf{e}'$.*

Proof. Subtracting the two openings gives $(\mathbf{s} - \mathbf{s}')^\top \mathbf{G} = (\mathbf{e}' - \mathbf{e})^\top \pmod{q}$. Let $\Delta := \mathbf{s} - \mathbf{s}'$. If $\Delta \neq \mathbf{0}$, then by the definition of $\delta_{\mathbf{G}}$, $\|\langle \Delta^\top \mathbf{G} \rangle_q\|_\infty \geq \delta_{\mathbf{G}}$. On the other hand, $\delta_{\mathbf{G}} \leq q/2$ implies $\|\mathbf{e}\|_\infty, \|\mathbf{e}'\|_\infty < q/4$, and hence every coordinate of $\mathbf{e}' - \mathbf{e}$ lies in $(-q/2, q/2)$. Its centred representative modulo q is therefore exactly the integer difference $\mathbf{e}' - \mathbf{e}$. Thus $\|\mathbf{e}' - \mathbf{e}\|_\infty \leq \|\mathbf{e}'\|_\infty + \|\mathbf{e}\|_\infty < \delta_{\mathbf{G}}$, a contradiction. Hence $\Delta = \mathbf{0}$, so $\mathbf{s} = \mathbf{s}'$, and then $\mathbf{e}' - \mathbf{e} \equiv \mathbf{0} \pmod{q}$. Since $\|\mathbf{e}' - \mathbf{e}\|_\infty < q/2$, this gives $\mathbf{e} = \mathbf{e}'$ over the integers.

Lemma 7 (Uniqueness of LWE Solutions). *Let q be prime, $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, and let $B \leq \beta q$ for a fixed constant $0 < \beta < 1/4$. Assume that $q^n(4\beta + 1/q)^m \leq \text{negl}(\lambda)$. Then, except with negligible probability over $\mathbf{A}$, every $\mathbf{b}^\top \in \mathbb{Z}_q^m$ has at most one opening $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top \pmod{q}$ with $\mathbf{s} \in \mathbb{Z}_q^n$, $\mathbf{e} \in \mathbb{Z}^m$, and $\|\mathbf{e}\|_\infty \leq B$.*

Proof. Fix any nonzero $\Delta \in \mathbb{Z}_q^n$. Since $\mathbf{A}$ is uniform, $\Delta^\top \mathbf{A}$ is uniform over $\mathbb{Z}_q^m$. Using centered representatives in $(-q/2, q/2]$, one coordinate lies in $[-2B, 2B]$ with probability at most $(4B + 1)/q \leq 4\beta + 1/q$. Hence

$$\Pr[\|\Delta^\top \mathbf{A}\|_\infty \leq 2B] \leq \left(4\beta + \frac{1}{q}\right)^m.$$

By a union bound over all nonzero $\Delta \in \mathbb{Z}_q^n$,

$$\Pr[\exists \Delta \neq \mathbf{0} : \|\Delta^\top \mathbf{A}\|_\infty \leq 2B] \leq q^n(4\beta + 1/q)^m \leq \text{negl}(\lambda).$$

So, except with negligible probability, every nonzero Δ satisfies $\|\Delta^\top \mathbf{A}\|_\infty > 2B$.

Now suppose $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top = \mathbf{s}'^\top \mathbf{A} + \mathbf{e}'^\top \pmod{q}$, where $\|\mathbf{e}\|_\infty, \|\mathbf{e}'\|_\infty \leq B$. Let $\Delta = \mathbf{s} - \mathbf{s}'$. Then $\Delta^\top \mathbf{A} = (\mathbf{e}' - \mathbf{e})^\top \pmod{q}$. Because $B \leq \beta q$ with $\beta < 1/4$, every coordinate of $\mathbf{e}' - \mathbf{e}$ lies in $(-q/2, q/2)$; hence its centred representative modulo q is the integer vector $\mathbf{e}' - \mathbf{e}$. If $\Delta \neq \mathbf{0}$, then $\|\Delta^\top \mathbf{A}\|_\infty > 2B$, while $\|\mathbf{e}' - \mathbf{e}\|_\infty \leq \|\mathbf{e}'\|_\infty + \|\mathbf{e}\|_\infty \leq 2B$, a contradiction. Thus $\Delta = \mathbf{0}$, so $\mathbf{s} = \mathbf{s}'$, and the same centred-representative argument gives $\mathbf{e} = \mathbf{e}'$ over the integers.

Following [16], [2] and the above two lemmas, we have the following.

Lemma 8 (Gadget LWE Inversion). *Let $\mathbf{G} \in \mathbb{Z}_q^{n \times w}$ be the gadget matrix. There is a p.p.t algorithm $\text{GInvert}(\mathbf{b}, \mathbf{G})$ with the following behaviour. On input $\mathbf{b} \in \mathbb{Z}_q^w$, it either returns a pair $(\mathbf{s}, \mathbf{e}) \in \mathbb{Z}_q^n \times \mathbb{Z}^w$ satisfying $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{G} + \mathbf{e}^\top \pmod{q}$ and $\|\mathbf{e}\|_\infty < \delta_{\mathbf{G}}/2$, if such a pair exists. By Lemma 6, this pair is unique. Otherwise, $\text{GInvert}(\mathbf{b}, \mathbf{G})$ returns $\perp$.*

Lemma 9 (Trapdoor LWE Inversion). *Let $\mathbf{F} \in \mathbb{Z}_q^{n \times N}$ and let $\mathbf{T}$ be a short trapdoor basis for $\Lambda_q^\perp(\mathbf{F})$, and let $B < q/4$ be a public decoding radius. There is a p.p.t algorithm $\text{Invert}(\mathbf{b}, \mathbf{F}, \mathbf{T})$ which, on input $\mathbf{b} \in \mathbb{Z}_q^N$, either returns a pair $(\mathbf{s}, \mathbf{e}) \in \mathbb{Z}_q^n \times \mathbb{Z}^N$ satisfying $\mathbf{b}^\top = \mathbf{s}^\top \mathbf{F} + \mathbf{e}^\top \pmod{q}$ and $\|\mathbf{e}\|_\infty < B$, if such a pair lies within the trapdoor inversion radius, or returns $\perp$ otherwise. In our application $B = B_{\text{LWE}}$; because the first block of $\mathbf{F}$ is the matrix $\mathbf{A}$ from Lemma 7, such an opening is unique except with negligible probability.*

Lemma 10 (ToBasis from [2,16]). *Let $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and $\mathbf{B} \in \mathbb{Z}_q^{n \times w}$, and set $\mathbf{F} = [\mathbf{A} \mid \mathbf{B}] \in \mathbb{Z}_q^{n \times N}$, where $N = m + w$. For a Gaussian parameter σ above the required smoothing bound, the standard left and right trapdoor-delegation algorithms give a p.p.t algorithm $\text{ToBasis}(\mathbf{A}, \text{td}, \mathbf{B}, \sigma)$. Here td may be either a trapdoor for $\mathbf{A}$, or a pair $(\mathbf{R}', \mathbf{H})$ such that $\mathbf{R}' \in \mathbb{Z}^{m \times w}$ is short, $\mathbf{H} \in \mathbb{Z}_q^{n \times n}$ is invertible, and $\mathbf{B} = \mathbf{A}\mathbf{R}' + \mathbf{H}\mathbf{G} \pmod{q}$. In either case, the algorithm outputs a short basis $\mathbf{T} \in \mathbb{Z}^{N \times N}$ of $\Lambda_q^\perp(\mathbf{F})$ with overwhelming probability. In particular, $\mathbf{F}\mathbf{T} = \mathbf{0} \pmod{q}$.*

More precisely, the randomised basis-delegation algorithms produce an output distribution statistically close to a canonical distribution $\mathcal{D}_{\text{Basis}}(\mathbf{F}, \sigma)$ that depends only on $\mathbf{F}$ and σ , and not on which of the two kinds of trapdoor information is used. The underlying Gaussian samples have distribution $D_{\Lambda_q^\perp(\mathbf{F}), \sigma}$, equivalently $D_{\mathbb{Z}^N, \sigma}$ conditioned on $\mathbf{F}\mathbf{x} = \mathbf{0} \pmod{q}$ by the conditional-Gaussian part of Lemma 3.

2.3 Full-Rank Difference Encoding

For a prime q and a degree- d irreducible polynomial over $\mathbb{Z}_q$, $\mathcal{R}_q := \mathbb{F}_q[X]/(f(X))$ is a finite field. Given an $\mathbb{Z}_q^d$ -element $\mathbf{a}$, we can uniquely identify a polynomial in $\mathcal{R}_q$ and vice versa, via coefficient embedding. We recall the notion of (extended) full-rank difference encoding (FRD) [2,8].

Definition 2 (Full-Rank Difference Encoding). *For a positive integer d and a prime q , a function $\phi : \mathcal{R}_q \rightarrow \mathbb{Z}_q^{d \times d}$ is a full-rank difference encoding if it is efficiently computable and satisfies 1) invertible difference, i.e., for $\mathbf{a}, \mathbf{b} \in \mathcal{R}_q$, $\mathbf{a} \neq \mathbf{b}$, $\phi(\mathbf{a}) - \phi(\mathbf{b})$ is invertible over $\mathbb{Z}_q$, and 2) homomorphism, i.e., for every $\mathbf{a}, \mathbf{b} \in \mathcal{R}_q$, $\phi(\mathbf{a} + \mathbf{b}) = \phi(\mathbf{a}) + \phi(\mathbf{b})$ and $\phi(\mathbf{a}\mathbf{b}) = \phi(\mathbf{a})\phi(\mathbf{b})$.*

Given $n > 0$ with $d \mid n$, define an extended FRD $H : \mathcal{R}_q \rightarrow \mathbb{Z}_q^{n \times n}$ as

$$H(\mathbf{a}) := \begin{bmatrix} \phi(\mathbf{a}) & 0 & \cdots & 0 \\ 0 & \phi(\mathbf{a}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \phi(\mathbf{a}) \end{bmatrix} \in \mathbb{Z}_q^{n \times n},$$

where n/d copies of $\phi(\mathbf{a})$ appear on the diagonal.

By construction, the extended FRD map H is itself an FRD and a ring homomorphism: for all $a, b \in \mathcal{R}_q$, $H(a + b) = H(a) + H(b)$ and $H(ab) = H(a)H(b)$. Efficient constructions of FRD and extended FRD encodings are given in [2]. As in [14], our construction uses the inner-product universal hash functions over extended FRD.

Lemma 11. *Let ℓ be a positive integer and let q be prime. Let $H : \mathcal{R}_q \rightarrow \mathbb{Z}_q^{n \times n}$ be the extended FRD map defined above. For each fixed $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_\ell) \in \mathcal{R}_q^\ell$, define*

$$f_{\mathbf{s}} : \mathcal{R}_q^\ell \rightarrow H(\mathcal{R}_q), \quad f_{\mathbf{s}}(\mathbf{x}_1, \dots, \mathbf{x}_\ell) = \sum_{i=1}^{\ell} H(\mathbf{x}_i)H(\mathbf{s}_i),$$

where $H(\mathcal{R}_q) = \{H(a) : a \in \mathcal{R}_q\}$. Since H is injective, $|H(\mathcal{R}_q)| = |\mathcal{R}_q| = q^d$. Then the collection $\mathcal{F} = \{f_{\mathbf{s}} : \mathbf{s} \in \mathcal{R}_q^\ell\}$, indexed by $\mathbf{s} \leftarrow U(\mathcal{R}_q^\ell)$, is universal. Namely, for all $(\mathbf{x}_1, \dots, \mathbf{x}_\ell) \neq (\mathbf{x}'_1, \dots, \mathbf{x}'_\ell)$,

$$\Pr_{\mathbf{s} \leftarrow U(\mathcal{R}_q^\ell)} [f_{\mathbf{s}}(\mathbf{x}_1, \dots, \mathbf{x}_\ell) = f_{\mathbf{s}}(\mathbf{x}'_1, \dots, \mathbf{x}'_\ell)] \leq q^{-d}.$$

Proof. Fix two distinct inputs $(\mathbf{x}_1, \dots, \mathbf{x}_\ell) \neq (\mathbf{x}'_1, \dots, \mathbf{x}'_\ell)$ and choose i^* such that $\mathbf{x}_{i^*} \neq \mathbf{x}'_{i^*}$. Since H is a ring homomorphism,

$$f_{\mathbf{s}}(\mathbf{x}_1, \dots, \mathbf{x}_\ell) - f_{\mathbf{s}}(\mathbf{x}'_1, \dots, \mathbf{x}'_\ell) = H\left(\sum_{i=1}^{\ell} (\mathbf{x}_i - \mathbf{x}'_i) \mathbf{s}_i\right).$$

Thus, a collision implies $\sum_i (\mathbf{x}_i - \mathbf{x}'_i) \mathbf{s}_i = \mathbf{0}$ in $\mathcal{R}_q$, since H is injective. Fix all $\mathbf{s}_i$ for $i \neq i^*$. Since $\mathcal{R}_q$ is a field and $\mathbf{x}_{i^*} - \mathbf{x}'_{i^*} \neq \mathbf{0}$, there is a unique value

$$\mathbf{s}_{i^*} = -(\mathbf{x}_{i^*} - \mathbf{x}'_{i^*})^{-1} \sum_{i \neq i^*} (\mathbf{x}_i - \mathbf{x}'_i) \mathbf{s}_i$$

that causes a collision. Since $\mathbf{s}_{i^*}$ is uniform over $\mathcal{R}_q$, the collision probability is upper bounded by $1/|\mathcal{R}_q| = q^{-d}$.

Lemma 12. *Let $H : \mathcal{R}_q \rightarrow \mathbb{Z}_q^{n \times n}$ be an extended FRD. For every nonzero $\mathbf{r} \in \mathbb{Z}_q^n$, the map $\mathbf{x} \mapsto H(\mathbf{x})^\top \mathbf{r}$ is injective. Moreover, there is an efficient algorithm $\text{RecoverId}(\mathbf{r}, \mathbf{z})$ that returns the unique $\mathbf{x} \in \mathcal{R}_q$ satisfying $\mathbf{z} = H(\mathbf{x})^\top \mathbf{r}$ if it exists, and returns $\perp$ otherwise.*

Proof. Suppose that for some $\mathbf{a}, \mathbf{b} \in \mathcal{R}_q$, $\mathbf{a} \neq \mathbf{b}$, $H(\mathbf{a})^\top \mathbf{r} = H(\mathbf{b})^\top \mathbf{r}$. Then, $(H(\mathbf{a}) - H(\mathbf{b}))^\top \mathbf{r} = \mathbf{0}$. By the full-rank difference property, $H(\mathbf{a}) - H(\mathbf{b}) = H(\mathbf{a} - \mathbf{b}) \in \text{GL}_n(\mathbb{Z}_q)$. Therefore $(H(\mathbf{a}) - H(\mathbf{b}))^\top$ is also invertible, and the above equation implies $\mathbf{r} = \mathbf{0}$, contradicting the assumption that $\mathbf{r} \neq \mathbf{0}$. Hence $\mathbf{a} = \mathbf{b}$, and the map is injective.

It remains to justify efficient recovery. Fix a basis $\{\mathbf{b}_1, \dots, \mathbf{b}_d\}$ of $\mathcal{R}_q$ over $\mathbb{F}_q$. Write $\mathbf{x} = \sum_{j=1}^d a_j \mathbf{b}_j$, $a_j \in \mathbb{Z}_q$. By linearity of H , $H(\mathbf{x})^\top \mathbf{r} = \sum_{j=1}^d a_j H(\mathbf{b}_j)^\top \mathbf{r} = \mathbf{z}$. Thus, given $(\mathbf{r}, \mathbf{z})$, recovering $\mathbf{x}$ reduces to solving the linear system $\mathbf{z} = \sum_{j=1}^d a_j H(\mathbf{b}_j)^\top \mathbf{r}$ over $\mathbb{Z}_q$ for the unknown coefficients $a_1, \dots, a_d$. Since the map is injective for $\mathbf{r} \neq \mathbf{0}$, the solution is unique whenever such an $\mathbf{x}$ exists; if the system is inconsistent, the algorithm returns $\perp$. So, RecoverId is efficient.

3 Identity-Based Encryption

Throughout, the identity space is $\text{ID}_\lambda = \mathcal{R}_q^\ell$. A complete identity is written $\text{id} = (\text{id}_1, \dots, \text{id}_\ell)$, where id_i denotes its i -th coordinate. When two or more complete identities occur in one experiment, we distinguish them by parenthesised superscripts, e.g., $\text{id}^{(0)}, \text{id}^{(1)}$, and write $\text{id}_i^{(b)}$ for coordinate i of $\text{id}^{(b)}$.

An identity-based encryption scheme is a tuple $\Pi = (\text{Setup}, \text{KeyGen}, \text{Enc}, \text{Dec})$ of p.p.t algorithms with identity space $\text{ID} = \{\text{ID}_\lambda\}_{\lambda \in \mathbb{N}}$, message space $\mathcal{M} = \{\mathcal{M}_\lambda\}_{\lambda \in \mathbb{N}}$, and ciphertext space $\mathcal{C} = \{\mathcal{C}_\lambda\}_{\lambda \in \mathbb{N}}$.

- $\text{Setup}(1^\lambda)$ outputs public parameters pp and a master secret key msk .
- $\text{KeyGen}(\text{pp}, \text{msk}, \text{id})$ takes an identity $\text{id} \in \text{ID}_\lambda$ and outputs a secret key sk_{id} .
- $\text{Enc}(\text{pp}, \text{id}, M)$ takes an identity $\text{id} \in \text{ID}_\lambda$ and a message $M \in \mathcal{M}_\lambda$, and outputs a ciphertext $C \in \mathcal{C}_\lambda$.
- $\text{Dec}(\text{pp}, \text{sk}_{\text{id}}, C)$ takes public parameters, a secret key, and a ciphertext, and outputs either a message M or $\perp$.

Correctness requires that for every $\lambda \in \mathbb{N}$, $\text{id} \in \text{ID}_\lambda$, and $M \in \mathcal{M}_\lambda$,

$$\Pr[\text{Dec}(\text{pp}, \text{KeyGen}(\text{pp}, \text{msk}, \text{id}), \text{Enc}(\text{pp}, \text{id}, M)) = M] \geq 1 - \text{negl}(\lambda),$$

where the probability is over the coin tosses of the algorithms.

We provide the three security definitions for anonymous IBE: (selective) ciphertext anonymity, function privacy and (strong) robustness. Anonymity requires that a ciphertext hide the message and recipient's identity.

Definition 3 (Selective Anonymity). *An AIBE scheme $\Pi = (\text{Setup}, \text{KeyGen}, \text{Enc}, \text{Dec})$ is selectively anonymous if for every p.p.t adversary $\mathcal{A}$, the advantage*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda) = \left| \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda) = 1] - 1/2 \right| \leq \text{negl}(\lambda)$$

where the experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda)$ is defined in Fig. 1.

Experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{ANO}-b}(\lambda)$:	Oracle $\mathcal{O}(\text{id})$:
1. $\text{id}^* \leftarrow \mathcal{A}(1^\lambda)$	1. $Q \leftarrow Q \cup \{\text{id}\}$
2. $(\text{pp}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$; $Q \leftarrow \emptyset$	2. Return $\text{KeyGen}(\text{pp}, \text{msk}, \text{id})$
3. $M^* \leftarrow \mathcal{A}^{\mathcal{O}(\cdot)}(\text{pp})$	
4. $C_0^* \leftarrow \text{Enc}(\text{pp}, \text{id}^*, M^*)$, $C_1^* \leftarrow U(\mathcal{C}_\lambda)$	
5. $b' \leftarrow \mathcal{A}^{\mathcal{O}(\cdot)}(C_b^*)$	
6. Return $(b' = b) \wedge (\text{id}^* \notin Q)$	

Fig. 1. Selective anonymity experiment for IBE.

Function privacy, formalised by Boneh et al. [6], requires that a secret key reveal no information about the identity for which it was generated except the absolute minimum necessary. We follow the original definition [6]. Let $X = (X_1, \dots, X_T)$ be a joint distribution over ID_λ^T . A (T, k) -block-source over ID_λ is a distribution such that, for every $i \in [T]$ and every fixing $x_1, \dots, x_{i-1}$ of the previous blocks, the conditional distribution $X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}$ has min-entropy at least k .

Definition 4 ((T, k)-Block-Source Function Privacy). *An adversary $\mathcal{A}$ is admissible if every query X to RoR^{FP} is a (T, k) -block-source over ID_λ . Define*

$$\text{Adv}_{\Pi, T, k, \mathcal{A}}^{\text{FP}}(\lambda) = \left| \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{FP}, \text{real}}(\lambda) = 1] - \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{FP}, \text{rand}}(\lambda) = 1] \right|.$$

Experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{FP}, \text{mode}}(\lambda)$	Oracle $\text{RoR}^{\text{FP}}(\text{mode}, \text{msk}, X)$
1. $(\text{pp}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$ 2. $b \leftarrow \mathcal{A}^{\text{RoR}^{\text{FP}}(\text{mode}, \text{msk}, \cdot), \mathcal{O}_{\text{KeyGen}}(\cdot)}(1^\lambda, \text{pp})$ 3. Return b Oracle $\mathcal{O}_{\text{KeyGen}}(\text{id})$ 1. Return $\text{KeyGen}(\text{pp}, \text{msk}, \text{id})$	1. If $\text{mode} = \text{real}$, $(\text{id}^{(1)}, \dots, \text{id}^{(T)}) \leftarrow X$; If $\text{mode} = \text{rand}$, $(\text{id}^{(1)}, \dots, \text{id}^{(T)}) \leftarrow U(\text{ID}_\lambda^T)$ 2. $\text{sk} \leftarrow (\text{KeyGen}(\text{pp}, \text{msk}, \text{id}^{(j)}))_{j \in [T]}$ 3. Return sk

Fig. 2. Real-or-random function privacy experiment for IBE.

We say that Π is (T, k) -block-source function-private if $\text{Adv}_{\Pi, T, k, \mathcal{A}}^{\text{FP}}(\lambda)$ is negligible for every admissible p.p.t adversary $\mathcal{A}$. It is statistically (T, k) -block-source function-private if the same holds against computationally unbounded admissible adversaries making polynomially many oracle queries.

Remark 1 (Multi-Shot Queries). The experiment allows the adversary to query RoR^{FP} polynomially many times. As in the source-based formulation of BRS [6], this multi-shot definition is polynomially equivalent to the single-shot version in which the adversary makes only one RoR^{FP} query.

Robustness requires that a ciphertext should not decrypt successfully under decryption keys associated with two distinct identities. We use the strong robustness notion of ABN10 [1], where the adversary is allowed to produce an arbitrary ciphertext, which implies the weak robustness notion [1].

Experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda)$	Oracle $\mathcal{O}_{\text{KeyGen}}(\text{id})$
1. $(\text{pp}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$; set $Q_{\text{KeyGen}} \leftarrow \emptyset$ 2. $(\text{id}^{(0)}, \text{id}^{(1)}, C^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{KeyGen}}(\cdot)}(\text{pp})$ 3. If $\text{id}^{(0)} = \text{id}^{(1)}$, return 0 4. If $\{\text{id}^{(0)}, \text{id}^{(1)}\} \cap Q_{\text{KeyGen}} \neq \emptyset$, return 0 5. $\text{sk}_b \leftarrow \text{KeyGen}(\text{pp}, \text{msk}, \text{id}^{(b)})$ for $b \in \{0, 1\}$ 6. Return 1 iff both $\text{Dec}(\text{pp}, \text{sk}_0, C^*) \neq \perp$, and $\text{Dec}(\text{pp}, \text{sk}_1, C^*) \neq \perp$	1. $Q_{\text{KeyGen}} \leftarrow Q_{\text{KeyGen}} \cup \{\text{id}\}$ 2. Return $\text{KeyGen}(\text{pp}, \text{msk}, \text{id})$

Fig. 3. Strong robustness experiment for IBE.

Definition 5 (Strong Robustness). We say an AIBE scheme Π SROB-CPA is secure if for every p.p.t adversary $\mathcal{A}$,

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda) = \Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda) = 1] \leq \text{negl}(\lambda)$$

where the experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda)$ is defined in Fig. 3.

4 Construction

The four algorithms of our AIBE construction are as follows. The input to KeyGen and Enc is one identity vector $\text{id} = (\text{id}_1, \dots, \text{id}_\ell) \in \mathcal{R}_q^\ell$, rather than ℓ distinct identities.

- Setup(1^λ):
 1. Sample $\bar{\mathbf{A}} \leftarrow U(\mathbb{Z}_q^{n \times (m-w)})$ and run $(\mathbf{A}, \mathbf{R}) \leftarrow \text{TrapGen}(\bar{\mathbf{A}}, \mathbf{I}_n, \sigma_{\text{td}})$.
 2. Sample $\mathbf{A}_i \leftarrow U(\mathbb{Z}_q^{n \times w})$ for every $i \in [\ell]$, and sample $\mathbf{u} \leftarrow U(\mathbb{Z}_q^n)$.
 3. Return $\text{msk} := \mathbf{R}$ and $\text{pp} := (\mathbf{A}, \mathbf{A}_1, \dots, \mathbf{A}_\ell, \mathbf{u})$.
- KeyGen($\text{pp}, \text{msk}, \text{id}$):
 1. Sample $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_\ell) \leftarrow U(\mathcal{R}_q^\ell)$ and set $\mathbf{Y} = \sum_{i=1}^\ell H(\text{id}_i)H(\mathbf{s}_i)$.
 2. For each $i \in [\ell]$, set $\mathbf{D}_i = \mathbf{G}^{-1}(H(\mathbf{s}_i)\mathbf{G})$. Compute $\mathbf{A}_s = \sum_i \mathbf{A}_i \mathbf{D}_i$, $\mathbf{B}_{s,Y} = \mathbf{A}_s + \mathbf{Y}\mathbf{G}$, and $\mathbf{F}_{s,Y} = [\mathbf{A} \mid \mathbf{B}_{s,Y}]$.
 3. Compute $\mathbf{T}_{s,Y} \leftarrow \text{ToBasis}(\mathbf{A}, \mathbf{R}, \mathbf{B}_{s,Y}, \sigma_{\text{del}})$.
 4. Output $\text{sk}_{\text{id}} = (\mathbf{s}, \mathbf{Y}, \mathbf{T}_{s,Y})$.
- Enc(pp, id, b):
 1. Sample $\mathbf{r} \leftarrow U(\mathbb{Z}_q^n \setminus \{\mathbf{0}\})$, $\mathbf{e}_0 \leftarrow D_{\mathbb{Z},s}^m$, $e \leftarrow D_{\mathbb{Z},s}$, and $\mathbf{R}_i \leftarrow U(\{-1, 1\}^{m \times w})$ for every $i \in [\ell]$.
 2. Compute $\mathbf{c}_0 = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_0 \pmod{q}$.
 3. For every $i \in [\ell]$, compute $\mathbf{c}_i = (\mathbf{A}_i + H(\text{id}_i)\mathbf{G})^\top \mathbf{r} + \mathbf{R}_i^\top \mathbf{e}_0 \pmod{q}$.
 4. Compute $c = \mathbf{u}^\top \mathbf{r} + e + b \lfloor q/2 \rfloor \pmod{q}$.
 5. Output $C = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_\ell, c)$.
- Dec(pp, sk, C):
 1. Parse $\text{sk} = (\mathbf{s}, \mathbf{Y}, \mathbf{T}_{s,Y})$ and $C = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_\ell, c)$.
 2. Set $\mathbf{A}_s = \sum_{i=1}^\ell \mathbf{A}_i \mathbf{G}^{-1}(H(\mathbf{s}_i)\mathbf{G})$, $\mathbf{B}_{s,Y} = \mathbf{A}_s + \mathbf{Y}\mathbf{G}$, and $\mathbf{F}_{s,Y} = [\mathbf{A} \mid \mathbf{B}_{s,Y}]$.
 3. Compute $\hat{\mathbf{c}}_s = \sum_{i=1}^\ell (\mathbf{G}^{-1}(H(\mathbf{s}_i)\mathbf{G}))^\top \mathbf{c}_i$ and set $\mathbf{c}_s = (\mathbf{c}_0, \hat{\mathbf{c}}_s)$.
 4. Run $(\hat{\mathbf{r}}, \hat{\mathbf{e}}_{\text{comb}}) \leftarrow \text{Invert}(\mathbf{c}_s, \mathbf{F}_{s,Y}, \mathbf{T}_{s,Y})$ using decoding radius B_{LWE} . If inversion fails or $\hat{\mathbf{r}} = \mathbf{0}$, return $\perp$.
 5. For every $i \in [\ell]$, compute $\mathbf{d}_i = \mathbf{c}_i - \mathbf{A}_i^\top \hat{\mathbf{r}} \pmod{q}$ and run $(\hat{\mathbf{z}}_i, \hat{\mathbf{e}}_i) \leftarrow \text{GInvert}(\mathbf{d}_i, \mathbf{G})$. If any gadget inversion fails, return $\perp$.
 6. For every $i \in [\ell]$, compute $\hat{\text{id}}_i \leftarrow \text{RecoverId}(\hat{\mathbf{r}}, \hat{\mathbf{z}}_i)$. If $\hat{\text{id}}_i = \perp$ or $\hat{\mathbf{z}}_i \neq H(\hat{\text{id}}_i)^\top \hat{\mathbf{r}}$, return $\perp$.
 7. If $\mathbf{Y} \neq \sum_{i=1}^\ell H(\hat{\text{id}}_i)H(\mathbf{s}_i)$, return $\perp$.
 8. Return 0, if $\rho = c - \mathbf{u}^\top \hat{\mathbf{r}} \pmod{q}$ closer to 0. Else, return 1.

Parameter selection. Let $n = \text{poly}(\lambda)$ and $\ell = \text{poly}(\lambda)$. Choose a prime modulus $q = \Theta(\lambda^\kappa)$ for a sufficiently large constant κ , and choose $d = \omega(1)$ with $d \mid n$ such that q^d is super-polynomial in λ . Let $w = n \lceil \log_2 q \rceil$, fix a constant $0 < \beta < 1/4$, and choose $m = \Theta(n \log q)$ such that $m > (n+2) \log q + \omega(\log n)$ and m is large enough for Lemmas 2, 3, and 7; in particular, require $q^n(4\beta + 1/q)^m \leq \text{negl}(\lambda)$. Let α be the LWE noise rate and set $s = \alpha q$, where $\alpha q \geq 2\sqrt{n}$. Choose B_{LWE} so that $\ell s m w \cdot \omega(\sqrt{\log \lambda}) < B_{\text{LWE}} \leq \beta q$, and require $s m \cdot \omega(\sqrt{\log \lambda}) < \delta_{\mathbf{G}}/2$ and $s \cdot \omega(\sqrt{\log \lambda}) < q/4$. Finally, choose $\sigma_{\text{td}} \geq \omega(\sqrt{\log m})$ and

$$\sigma_{\text{del}} = \omega(\sqrt{\log(m+w)}) \max\{\sigma_{\text{td}}\sqrt{m} \cdot \omega(\sqrt{\log m}), \ell w \sqrt{m}\}.$$

These choices ensure that every call to ToBasis is valid and that Qq^{-d} and the Gaussian-tail failure probabilities are negligible for every polynomial Q .

Correctness. Let $\mathbf{F}_{\mathbf{s}, \mathbf{Y}} := [\mathbf{A} \mid \mathbf{B}_{\mathbf{s}, \mathbf{Y}}]$. By Lemma 10, $\mathbf{T}_{\mathbf{s}, \mathbf{Y}}$ is a short basis of $\Lambda_q^\perp(\mathbf{F}_{\mathbf{s}, \mathbf{Y}})$ and $\mathbf{F}_{\mathbf{s}, \mathbf{Y}} \mathbf{T}_{\mathbf{s}, \mathbf{Y}} = \mathbf{0} \pmod{q}$. Encryption samples $\mathbf{r} \leftarrow U(\mathbb{Z}_q^n \setminus \{\mathbf{0}\})$, $\mathbf{e}_0 \leftarrow D_{\mathbb{Z}, s}^m$, $e \leftarrow D_{\mathbb{Z}, s}$, and $\mathbf{R}_i \leftarrow U(\{-1, 1\}^{m \times w})$. Hence $\mathbf{c}_0 = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_0$ and $\mathbf{c}_i = (\mathbf{A}_i + H(\text{id}_i) \mathbf{G})^\top \mathbf{r} + \mathbf{R}_i^\top \mathbf{e}_0 \pmod{q}$. Let $\mathbf{e}_i := \mathbf{R}_i^\top \mathbf{e}_0$. Since $\mathbf{G} \mathbf{D}_i = H(\mathbf{s}_i) \mathbf{G}$ and $\mathbf{Y} = \sum_i H(\text{id}_i) H(\mathbf{s}_i)$, decryption computes

$$\begin{aligned} \tilde{\mathbf{c}}_{\mathbf{s}} &= \sum_{i=1}^{\ell} \mathbf{D}_i^\top ((\mathbf{A}_i + H(\text{id}_i) \mathbf{G})^\top \mathbf{r} + \mathbf{e}_i) \\ &= \left(\mathbf{A}_{\mathbf{s}} + \sum_{i=1}^{\ell} H(\text{id}_i) H(\mathbf{s}_i) \mathbf{G} \right)^\top \mathbf{r} + \sum_{i=1}^{\ell} \mathbf{D}_i^\top \mathbf{e}_i \\ &= \mathbf{B}_{\mathbf{s}, \mathbf{Y}}^\top \mathbf{r} + \tilde{\mathbf{e}}_{\mathbf{s}} \pmod{q}, \end{aligned}$$

where $\tilde{\mathbf{e}}_{\mathbf{s}} := \sum_i \mathbf{D}_i^\top \mathbf{e}_i$. Therefore $\mathbf{c}_{\mathbf{s}} := (\mathbf{c}_0, \tilde{\mathbf{c}}_{\mathbf{s}}) = \mathbf{F}_{\mathbf{s}, \mathbf{Y}}^\top \mathbf{r} + (\mathbf{e}_0, \tilde{\mathbf{e}}_{\mathbf{s}}) \pmod{q}$.

It remains to bound the error. For a one-dimensional discrete Gaussian sample $z \leftarrow D_{\mathbb{Z}, s}$, taking the tail threshold in $s \cdot \omega(\sqrt{\log \lambda})$ gives failure probability $\exp(-\omega(\log \lambda)) = \lambda^{-\omega(1)}$. Since $m = \text{poly}(\lambda)$, a union bound over all m coordinates still gives a negligible failure probability. Hence $\|\mathbf{e}_0\|_2 \leq s\sqrt{m} \cdot \omega(\sqrt{\log \lambda})$ and $|e| \leq s \cdot \omega(\sqrt{\log \lambda})$ with overwhelming probability. Since every column of $\mathbf{R}_i$ has norm $\sqrt{m}$, $\|\mathbf{e}_i\|_\infty = \|\mathbf{R}_i^\top \mathbf{e}_0\|_\infty \leq \sqrt{m} \|\mathbf{e}_0\|_2 \leq sm \cdot \omega(\sqrt{\log \lambda})$. Moreover, every column of $\mathbf{D}_i$ has ℓ_1 -norm at most w , so $\|\mathbf{D}_i^\top \mathbf{e}_i\|_\infty \leq w \|\mathbf{e}_i\|_\infty \leq smw \cdot \omega(\sqrt{\log \lambda})$. Hence $\|(\mathbf{e}_0, \tilde{\mathbf{e}}_{\mathbf{s}})\|_\infty \leq \ell smw \cdot \omega(\sqrt{\log \lambda}) < B_{\text{LWE}} < q/4$. Thus inversion using $\mathbf{T}_{\mathbf{s}, \mathbf{Y}}$ recovers the correct linear term $\mathbf{r}$. For each gadget residual $\mathbf{d}_i = \mathbf{c}_i - \mathbf{A}_i^\top \mathbf{r} \pmod{q}$, we have $\mathbf{d}_i = \mathbf{G}^\top H(\text{id}_i)^\top \mathbf{r} + \mathbf{e}_i \pmod{q}$ and $\|\mathbf{e}_i\|_\infty \leq sm \cdot \omega(\sqrt{\log \lambda}) < \delta_{\mathbf{G}}/2$. Hence $\mathbf{G} \text{Invert}$ recovers the unique value $H(\text{id}_i)^\top \mathbf{r}$. Finally, since $c = \mathbf{u}^\top \mathbf{r} + e + \lfloor q/2 \rfloor b \pmod{q}$ and $|e| \leq s \cdot \omega(\sqrt{\log \lambda}) < q/4$, rounding recovers b correctly.

5 Security Analysis

5.1 Selective Anonymity

Theorem 1 (Selective anonymity from LWE). *Let Π be the AIBE scheme described above, with parameters chosen as above. For every p.p.t adversary $\mathcal{A}$ against the anonymous data privacy of Π , there exists a p.p.t distinguisher $\mathcal{B}$ for $\text{LWE}_{n, m+1, q, D_{\mathbb{Z}, s}}$ such that $\text{Adv}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n, m+1, q, D_{\mathbb{Z}, s}}}(\lambda) + \text{negl}(\lambda)$.*

Proof. Let $\mathcal{A}$ be a p.p.t adversary in the selective anonymity experiment, where the challenge identity $\text{id}^* = (\text{id}_1^*, \dots, \text{id}_\ell^*)$ is fixed before setup. Let a polynomial Q be the number of key generation queries made by $\mathcal{A}$. We prove the theorem by a sequence of games where a well-defined binary value is output at the end of each game. Let $G_i = 1$ denote the event that the i -th game returns 1.

Game G_0 . This is the real experiment. The challenger runs $(\text{pp}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$, answers key generation queries using KeyGen , and gives $\mathcal{A}$ the real challenge ciphertext $C^* \leftarrow \text{Enc}(\text{pp}, \text{id}^*, M^*)$.

Game G_1 . This game is identical to **Game G_0** , except that the public matrices are generated in simulated form. The simulator samples $\bar{\mathbf{A}} \leftarrow U(\mathbb{Z}_q^{n \times (m-w)})$ and runs $(\mathbf{A}, \mathbf{R}) \leftarrow \text{TrapGen}(\bar{\mathbf{A}}, \mathbf{I}_n, \sigma_{\text{td}})$. For each $i \in [\ell]$, it samples a proof-only matrix $\mathbf{R}_i^* \leftarrow U(\{-1, 1\}^{m \times w})$ and sets $\mathbf{A}_i = \mathbf{A}\mathbf{R}_i^* - H(\text{id}_i^*)\mathbf{G} \pmod{q}$.

Let $\mathbf{R}^* = [\mathbf{R}_1^*] \cdots [\mathbf{R}_\ell^*]$. By Lemma 4, the distribution $(\mathbf{A}, \mathbf{A}\mathbf{R}^*)$ is statistically close to $(\mathbf{A}, \mathbf{U})$ for uniform $\mathbf{U} \leftarrow U(\mathbb{Z}_q^{n \times \ell w})$. Hence $(\mathbf{A}_1, \dots, \mathbf{A}_\ell)$ is statistically close to a tuple of independent uniform public matrices, since subtracting the fixed matrices $H(\text{id}_i^*)\mathbf{G}$ preserves uniformity.

Key generation queries are answered using the real master trapdoor. On input $\text{id} \neq \text{id}^*$, the challenger samples $\mathbf{s} \leftarrow U(\mathcal{R}_q^\ell)$, sets $\mathbf{Y} = \sum_i H(\text{id}_i)H(\mathbf{s}_i)$, $\mathbf{D}_i = \mathbf{G}^{-1}(H(\mathbf{s}_i)\mathbf{G})$, $\mathbf{A}_\mathbf{s} = \sum_i \mathbf{A}_i\mathbf{D}_i$, $\mathbf{B}_{\mathbf{s}, \mathbf{Y}} = \mathbf{A}_\mathbf{s} + \mathbf{Y}\mathbf{G}$, and $\mathbf{F}_{\mathbf{s}, \mathbf{Y}} = [\mathbf{A} \mid \mathbf{B}_{\mathbf{s}, \mathbf{Y}}]$. It computes $\mathbf{T}_{\mathbf{s}, \mathbf{Y}} \leftarrow \text{ToBasis}(\mathbf{A}, \mathbf{R}, \mathbf{B}_{\mathbf{s}, \mathbf{Y}}, \sigma_{\text{del}})$ and returns $\text{sk}_{\text{id}} = (\mathbf{s}, \mathbf{Y}, \mathbf{T}_{\mathbf{s}, \mathbf{Y}})$. By Lemma 10, this is exactly the real key generation procedure.

So,

$$|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{negl}(\lambda).$$

Game G_2 . This game is identical to **Game G_1** , except that key queries are answered without the master trapdoor. The simulator samples $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$ and, for every $i \in [\ell]$, samples $\mathbf{R}_i^* \xleftarrow{\$} \{-1, 1\}^{m \times w}$ and sets $\mathbf{A}_i = \mathbf{A}\mathbf{R}_i^* - H(\text{id}_i^*)\mathbf{G} \pmod{q}$. By Lemma 2, the real $\mathbf{A}$ is statistically close to uniform. By Lemma 4, $(\mathbf{A}, \mathbf{A}\mathbf{R}_1^*, \dots, \mathbf{A}\mathbf{R}_\ell^*)$ is statistically close to $(\mathbf{A}, \mathbf{U}_1, \dots, \mathbf{U}_\ell)$, where each $\mathbf{U}_i$ is uniform over $\mathbb{Z}_q^{n \times w}$. Since subtracting the fixed matrices $H(\text{id}_i^*)\mathbf{G}$ preserves uniformity, the simulated public parameters are statistically close to those in **Game G_1** .

For a key query $\text{id} = (\text{id}_1, \dots, \text{id}_\ell) \neq \text{id}^*$, sample $\mathbf{s} \xleftarrow{\$} \mathcal{R}_q^\ell$, set $\mathbf{Y} = \sum_i H(\text{id}_i)H(\mathbf{s}_i)$ and $\mathbf{D}_i = \mathbf{G}^{-1}(H(\mathbf{s}_i)\mathbf{G})$, and define $\mathbf{A}_\mathbf{s} = \sum_i \mathbf{A}_i\mathbf{D}_i$, $\mathbf{B}_{\mathbf{s}, \mathbf{Y}} = \mathbf{A}_\mathbf{s} + \mathbf{Y}\mathbf{G}$, $\mathbf{F}_{\mathbf{s}, \mathbf{Y}} = [\mathbf{A} \mid \mathbf{B}_{\mathbf{s}, \mathbf{Y}}]$, $\mathbf{R}_\mathbf{s}^* = \sum_i \mathbf{R}_i^*\mathbf{D}_i$, and $\Delta_{\text{id}, \mathbf{s}} = \sum_i (H(\text{id}_i) - H(\text{id}_i^*))H(\mathbf{s}_i)$. Since $\mathbf{G}\mathbf{D}_i = H(\mathbf{s}_i)\mathbf{G}$, we have $\mathbf{B}_{\mathbf{s}, \mathbf{Y}} = \mathbf{A}\mathbf{R}_\mathbf{s}^* + \Delta_{\text{id}, \mathbf{s}}\mathbf{G} \pmod{q}$. Since H is a ring homomorphism, $\Delta_{\text{id}, \mathbf{s}} = H(\sum_i (\text{id}_i - \text{id}_i^*)\mathbf{s}_i)$. Since $\text{id} \neq \text{id}^*$, Lemma 11 gives $\Pr[\Delta_{\text{id}, \mathbf{s}} \text{ is not invertible}] \leq q^{-d}$; if this event occurs, the simulator aborts. Moreover, each column of $\mathbf{D}_i$ has ℓ_1 -norm at most w , and each column of $\mathbf{R}_i^*$ has Euclidean norm $\sqrt{m}$, so $\|\mathbf{R}_\mathbf{s}^*\| \leq \ell w \sqrt{m}$. By the parameter choice above, σ_{del} is large enough to invoke the right-delegation instantiation of Lemma 10.

The simulator samples $\mathbf{T}_{\mathbf{s}, \mathbf{Y}} \leftarrow \text{ToBasis}(\mathbf{A}, \mathbf{R}_\mathbf{s}^*, \mathbf{B}_{\mathbf{s}, \mathbf{Y}}, \sigma_{\text{del}})$ using the right-delegation instantiation, and returns $\text{sk}_{\text{id}} = (\mathbf{s}, \mathbf{Y}, \mathbf{T}_{\mathbf{s}, \mathbf{Y}})$. By Lemma 10, the basis generated in **Game G_1** and the basis generated in **Game G_2** are both statistically close to $\mathcal{D}_{\text{Basis}}(\mathbf{F}_{\mathbf{s}, \mathbf{Y}}, \sigma_{\text{del}})$. This canonical distribution depends only on $\mathbf{F}_{\mathbf{s}, \mathbf{Y}} = [\mathbf{A} \mid \mathbf{B}_{\mathbf{s}, \mathbf{Y}}]$ and σ_{del} , not on whether the sampler is implemented using the real trapdoor $\mathbf{R}$ or the simulated $\mathbf{R}_\mathbf{s}^*$. Hence the returned secret keys are statistically close, except for the abort event. Over at most Q key queries, the abort probability is at most Qq^{-d} . Therefore,

$$|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq Qq^{-d} + \text{negl}(\lambda).$$

Game G_3 . This game is identical to **Game G_2** , except that the challenge ciphertext is replaced by a uniformly random element of the ciphertext space.

We show that **Game** G_2 and **Game** G_3 are indistinguishable by an LWE reduction. The reduction receives a challenge $(\mathbf{A}, \mathbf{u}, \mathbf{b}_0, b_u)$, where either $\mathbf{b}_0 = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_0$ and $b_u = \mathbf{u}^\top \mathbf{r} + e$, or $\mathbf{b}_0 \leftarrow U(\mathbb{Z}_q^m)$ and $b_u \leftarrow U(\mathbb{Z}_q)$ are uniform. As in **Game** G_2 , it samples hidden matrices $\mathbf{R}_1^*, \dots, \mathbf{R}_\ell^*$ and sets $\mathbf{A}_i = \mathbf{A}\mathbf{R}_i^* - H(\text{id}_i^*)\mathbf{G}$ for every $i \in [\ell]$. Key queries are answered as in **Game** G_2 . For the challenge message M^* , it returns $\mathbf{c}_0^* = \mathbf{b}_0$, $\mathbf{c}_i^* = \mathbf{R}_i^{*\top} \mathbf{b}_0$ for $i \in [\ell]$, $c^* = b_u + M^* \lfloor q/2 \rfloor$.

When the LWE instance is sampled from the LWE distribution, we have $\mathbf{b}_0 = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_0$ and $b_u = \mathbf{u}^\top \mathbf{r} + e$. Hence, for every $i \in [\ell]$,

$$\mathbf{c}_i^* = \mathbf{R}_i^{*\top} (\mathbf{A}^\top \mathbf{r} + \mathbf{e}_0) = (\mathbf{A}\mathbf{R}_i^*)^\top \mathbf{r} + \mathbf{R}_i^{*\top} \mathbf{e}_0 = (\mathbf{A}_i + H(\text{id}_i^*)\mathbf{G})^\top \mathbf{r} + \mathbf{R}_i^{*\top} \mathbf{e}_0.$$

In this case, the challenge ciphertext has the same distribution as in **Game** G_2 , up to the statistical distance already incurred in the simulation.

When the LWE instance is uniformly random, $\mathbf{b}_0$ and b_u are uniform. The last component is uniform since adding the fixed value $M^* \lfloor q/2 \rfloor$ preserves uniformity. Let $\mathbf{R}^* = [\mathbf{R}_1^* | \dots | \mathbf{R}_\ell^*]$. Lemma 5, with $K = \ell w$ and $\mathbf{v} = \mathbf{b}_0$, gives

$$(\mathbf{A}, \mathbf{b}_0, \mathbf{A}\mathbf{R}^*, \mathbf{R}^{*\top} \mathbf{b}_0) \approx_s (\mathbf{A}, \mathbf{b}_0, \mathbf{U}, \mathbf{z}),$$

where $\mathbf{U}$ and $\mathbf{z}$ are independent and uniform. The public blocks $\mathbf{A}_i = \mathbf{A}\mathbf{R}_i^* - H(\text{id}_i^*)\mathbf{G}$ and challenge blocks $\mathbf{c}_i^* = \mathbf{R}_i^{*\top} \mathbf{b}_0$ are deterministic affine post-processing of this tuple; hence their joint distribution, including $\mathbf{c}_0^* = \mathbf{b}_0$, is statistically close to independent uniform blocks. Further, by Lemma 10, conditioned on the public matrices and each query pair $(\mathbf{s}, \mathbf{Y})$, the returned basis is close to the canonical distribution $\mathcal{D}_{\text{Basis}}(\mathbf{F}_s, \mathbf{Y}, \sigma_{\text{del}})$ and therefore does not reveal the hidden $\mathbf{R}^*$. Including the adaptive key-query transcript is thus randomised post-processing and cannot increase statistical distance. So the entire challenge view is statistically close to **Game** G_3 , and

$$|\Pr[G_2 = 1] - \Pr[G_3 = 1]| \leq \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n, m+1, q, D_{\mathbb{Z}}, s}}(\lambda) + \text{negl}(\lambda).$$

In **Game** G_3 , the challenge ciphertext is uniform and independent of the experiment bit. Hence $\Pr[G_3(\lambda) = 1] = 1/2$. Combining the above bounds,

$$\begin{aligned} \text{Adv}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda) &\leq \sum_{i=0}^2 |\Pr[G_i = 1] - \Pr[G_{i+1} = 1]| \\ &\leq \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n, m+1, q, D_{\mathbb{Z}}, s}}(\lambda) + Qq^{-d} + \text{negl}(\lambda). \end{aligned}$$

As Q is polynomial in λ and q^d is super-polynomial in λ , Qq^{-d} is negligible. Therefore $\text{Adv}_{\Pi, \mathcal{A}}^{\text{ANO}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{\text{LWE}_{n, m+1, q, D_{\mathbb{Z}}, s}}(\lambda) + \text{negl}(\lambda)$ completes the proof.

5.2 Function Privacy

Theorem 2 (Function privacy). *Let Π be the IBE scheme described above. Suppose every query submitted to RoR^{FP} is a (T, k) -block-source over $\mathcal{R}_q^\ell$, where $k \geq \log |\mathcal{R}_q| + 2 \log(1/\varepsilon) + \Theta(1) = d \log q + 2 \log(1/\varepsilon) + \Theta(1)$. Then Π is statistically (T, k) -block-source function-private. More precisely, if the adversary receives in total N secret keys from RoR^{FP} , then*

$$\text{Adv}_{\Pi, T, k, \mathcal{A}}^{\text{FP}}(\lambda) \leq 2N\varepsilon + \text{negl}(\lambda).$$

Proof. By Remark 1, we focus on analysing a single query to RoR^{FP} . The ordinary oracle $\mathcal{O}_{\text{KeyGen}}$ is identical in the real and random experiments, so we only compare the secret keys returned by this RoR^{FP} query.

Let $X = (X_1, \dots, X_T)$ be the distribution submitted to RoR^{FP} . By admissibility, X is a (T, k) -block-source over $\mathcal{R}_q^\ell$. In the real experiment, the oracle samples $(\text{id}^{(1)}, \dots, \text{id}^{(T)}) \leftarrow X$. In the random experiment, it samples independent uniform identities $U_1, \dots, U_T \leftarrow U(\mathcal{R}_q^\ell)$.

For each $j \in [T]$, key generation samples a fresh independent seed $\mathbf{s}^{(j)} = (\mathbf{s}_1^{(j)}, \dots, \mathbf{s}_\ell^{(j)}) \leftarrow U(\mathcal{R}_q^\ell)$ and computes $\mathbf{Y}_j = \sum_{i=1}^\ell H(\text{id}_i^{(j)})H(\mathbf{s}_i^{(j)})$ in the real experiment, and analogously $\mathbf{Y}_j = \sum_{i=1}^\ell H(U_{j,i})H(\mathbf{s}_i^{(j)})$ in the random experiment. Define $\text{View}_{\text{mode}} = (\mathbf{s}^{(1)}, \mathbf{Y}_1, \dots, \mathbf{s}^{(T)}, \mathbf{Y}_T)$, where $\text{mode} \in \{\text{real}, \text{rand}\}$ determines whether the identities are sampled from X or uniformly from $\mathcal{R}_q^\ell$.

By Lemma 11, the family $f_{\mathbf{s}}$ defined by $f_{\mathbf{s}}(\text{id}) = \sum_{i=1}^\ell H(\text{id}_i)H(\mathbf{s}_i)$ is universal with collision probability at most q^{-d} . Although $f_{\mathbf{s}}(\text{id})$ is represented as an $n \times n$ matrix, since H is a ring homomorphism, it equals $H(\sum_i \text{id}_i \mathbf{s}_i)$, so its effective range has size $|\mathcal{R}_q| = q^d$. Applying Lemma 1 with domain $\mathcal{R}_q^\ell$ and range of size q^d , the distribution $\text{View}_{\text{real}}$ is εT -close to the corresponding ideal distribution. The same argument applies to $\text{View}_{\text{rand}}$, since independent uniform identities over $\mathcal{R}_q^\ell$ also form a (T, k) -block-source. Hence $\text{View}_{\text{real}}$ and $\text{View}_{\text{rand}}$ are within statistical distance at most $2\varepsilon T$.

It remains to relate this compressed view to the actual secret keys. Once $(\mathbf{s}^{(j)}, \mathbf{Y}_j)$ is fixed, KeyGen computes $\mathbf{A}_{\mathbf{s}^{(j)}}$, forms $\mathbf{F}_{\mathbf{s}^{(j)}, \mathbf{Y}_j} = [\mathbf{A} \mid \mathbf{A}_{\mathbf{s}^{(j)}} + \mathbf{Y}_j \mathbf{G}]$, and samples $\mathbf{T}_{\mathbf{s}^{(j)}, \mathbf{Y}_j}$ using $\text{ToBasis}(\mathbf{A}, \mathbf{R}, \mathbf{A}_{\mathbf{s}^{(j)}} + \mathbf{Y}_j \mathbf{G}, \sigma_{\text{del}})$. By Lemma 10, for fixed public parameters, the output distribution is statistically close to the canonical distribution $\mathcal{D}_{\text{Basis}}(\mathbf{F}_{\mathbf{s}^{(j)}, \mathbf{Y}_j}, \sigma_{\text{del}})$, which depends only on $\mathbf{F}_{\mathbf{s}^{(j)}, \mathbf{Y}_j}$ and σ_{del} , and hence only on $(\mathbf{s}^{(j)}, \mathbf{Y}_j)$ and the public parameters. Therefore, passing from the compressed view $(\mathbf{s}^{(j)}, \mathbf{Y}_j)$ to the actual secret key is randomised post-processing.

Since statistical distance cannot increase under randomised post-processing, the vectors of secret keys returned by this single RoR^{FP} query in the real and random experiments are within distance $2\varepsilon T + \text{negl}(\lambda)$. Finally, applying the single-shot to multi-shot hybrid over all RoR^{FP} outputs gives $\text{Adv}_{\Pi, T, k, \mathcal{A}}^{\text{FP}}(\lambda) \leq 2N\varepsilon + \text{negl}(\lambda)$, where N is the total number of secret keys returned by RoR^{FP} . This proves statistical (T, k) -block-source function privacy.

5.3 Strong Robustness

Theorem 3 (Strong robustness). *The scheme is SROB-CPA secure. More precisely, for every SROB-CPA adversary $\mathcal{A}$, $\text{Adv}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda) \leq q^{-d} + \text{negl}(\lambda)$.*

Proof. Consider the SROB-CPA experiment. The adversary outputs two distinct identities $\text{id}^{(0)} \neq \text{id}^{(1)}$ and an arbitrary ciphertext $C^* = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_\ell, c)$. The target keys are generated only after $(\text{id}^{(0)}, \text{id}^{(1)}, C^*)$ is fixed, so their seeds are fresh and independent of $\mathcal{A}$'s view. For each $b \in \{0, 1\}$, write $\text{sk}_b = (\mathbf{s}_b, \mathbf{Y}_b, \mathbf{T}_{\mathbf{s}_b, \mathbf{Y}_b})$.

Since decryption accepts under $\mathbf{sk}_b$, the call $\text{Invert}(\mathbf{c}_{s_b}, \mathbf{F}_{s_b, \mathbf{Y}_b}, \mathbf{T}_{s_b, \mathbf{Y}_b})$ returns a nonzero vector $\mathbf{r}_b$ and an error vector whose first block gives $\mathbf{c}_0 = \mathbf{A}^\top \mathbf{r}_b + \mathbf{e}_{b,0} \pmod{q}$ with $\|\mathbf{e}_{b,0}\|_\infty < B_{\text{LWE}} < q/4$.

Since the same base block $\mathbf{c}_0$ has two valid short LWE openings, Lemma 7 gives $\mathbf{r}_0 = \mathbf{r}_1$ except with negligible probability. Denote the common nonzero vector by $\mathbf{r}$.

For every $i \in [\ell]$, both decryptions compute the same residual $\mathbf{d}_i = \mathbf{c}_i - \mathbf{A}_i^\top \mathbf{r}$. Since both decryptions accept, each residual admits a valid gadget opening with error norm below $\delta_{\mathbf{G}}/2$. By Lemma 6, this opening is unique. Hence both decryptions recover the same value $\hat{\mathbf{z}}_i$ and the same identity component $\hat{\mathbf{id}}_i = \text{RecoverId}(\mathbf{r}, \hat{\mathbf{z}}_i)$, except with negligible probability. Thus both accepting decryptions recover the same claimed identity $\hat{\mathbf{id}} = (\hat{\mathbf{id}}_1, \dots, \hat{\mathbf{id}}_\ell)$. On this good event, $\hat{\mathbf{id}}$ is determined by C^* and the public parameters, independently of the fresh target seeds.

Acceptance under $\mathbf{sk}_b$ also means that the final consistency check passes, namely $\mathbf{Y}_b = \sum_{i=1}^\ell H(\hat{\mathbf{id}}_i)H(\mathbf{s}_{b,i})$. By key generation, $\mathbf{Y}_b = \sum_{i=1}^\ell H(\mathbf{id}_i^{(b)})H(\mathbf{s}_{b,i})$. Therefore $\sum_{i=1}^\ell H(\mathbf{id}_i^{(b)})H(\mathbf{s}_{b,i}) = \sum_{i=1}^\ell H(\hat{\mathbf{id}}_i)H(\mathbf{s}_{b,i})$ for both $b \in \{0, 1\}$.

Since $\mathbf{id}^{(0)} \neq \mathbf{id}^{(1)}$, the recovered identity $\hat{\mathbf{id}}$ cannot be equal to both identities. Let b^* be the first index in $\{0, 1\}$ such that $\mathbf{id}^{(b^*)} \neq \hat{\mathbf{id}}$. Then the accepting event implies the non-trivial collision $\sum_{i=1}^\ell H(\mathbf{id}_i^{(b^*)})H(\mathbf{s}_{b^*,i}) = \sum_{i=1}^\ell H(\hat{\mathbf{id}}_i)H(\mathbf{s}_{b^*,i})$, except with negligible probability.

Fix the public parameters, the adversary's transcript, the ciphertext C^* , and the recovered identity $\hat{\mathbf{id}}$. The seed $\mathbf{s}_{b^*}$ is freshly sampled from $U(\mathcal{R}_q^\ell)$ in the generation of $\mathbf{sk}_{b^*}$. Since $\mathbf{id}^{(b^*)} \neq \hat{\mathbf{id}}$, Lemma 11 gives

$$\Pr_{\mathbf{s}_{b^*} \leftarrow U(\mathcal{R}_q^\ell)} \left[\sum_i H(\mathbf{id}_i^{(b^*)})H(\mathbf{s}_{b^*,i}) = \sum_i H(\hat{\mathbf{id}}_i)H(\mathbf{s}_{b^*,i}) \right] \leq q^{-d}$$

Hence, $\Pr[\text{SROB}] \leq q^{-d} + \text{negl}(\lambda)$, and therefore $\text{Adv}_{\Pi, \mathcal{A}}^{\text{SROB-CPA}}(\lambda) \leq q^{-d} + \text{negl}(\lambda)$. Since q^d is super-polynomial by parameter selection, the bound is negligible. This proves SROB-CPA security.

6 Conclusion

We constructed a standard-model LWE-based function-private anonymous IBE scheme that satisfies strong robustness without applying the generic ABN transformations. The construction uses an ABB-style lattice IBE structure, but replaces the external robustness layer with an internal identity-consistency check. A secret key stores a universal-hash compression of the identity, and decryption accepts only when the identity recovered from the ciphertext openings is consistent with this compressed value. This makes the rejection test compatible with function privacy, since the original identity is not stored in the secret key.

The security analysis shows that anonymity follows from LWE, while statistical function privacy follows from the leftover-hash lemma for high-min-entropy identity sources. Strong robustness follows from the uniqueness of the

relevant LWE and gadget openings and from the collision bound of the identity-compression hash. Compared with the BRS13 standard-model LWE construction, our scheme removes the $\mu = \omega(\log \lambda)$ repetition layer and therefore reduces the identity-dependent public parameters, ciphertext blocks, and secret-key seed components. This gives a direct lattice-based answer to the robust function-private anonymous IBE problem left open by BRS13.

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